

A Potential Solution to Insufficient Target-Domain Noise Data: Transfer Learning and Noise Modeling

Xintong Dong^{ID}, Ming Cheng^{ID}, *Graduate Student Member, IEEE*, Hongzhou Wang, Guanghui Li, Jun Lin^{ID}, and Shaoping Lu^{ID}

Abstract—Recently, a number of deep learning (DL) methods are developed to attenuate the noise in seismic data. Most of them show good performance under a common precondition: the training and testing data are drawn from the same distribution. However, it is challenging to acquire sufficient noise training data whose distribution is the same as the noise presented in the testing seismic data; we call such noise data with the same distribution as target-domain noise. To address this issue, we propose a promising DL paradigm for seismic data denoising based on transfer learning and seismic noise modeling. We first utilize a Green-function-based modeling method for seismic noise to generate a massive amount of synthetic noise, which is similar to real seismic noise. Second, the high-authenticity synthetic noise is used as the pretraining data in the source domain. Finally, we utilize limited real target-domain noise data to fine-tune the partial trainable parameters of pretrained model, thus transferring it into the target domain. This proposed DL paradigm gets rid of the need for enough target-domain noise data, so as to extend the application scope of DL-based method in seismic data denoising. Moreover, we design a novel network architecture based on multicascade structure and attention mechanism. This DL paradigm shows extremely similar denoising performance to that of using a large amount of target-domain noise data in both synthetic and real examples, demonstrating its potential in mitigating the dependence of DL-based seismic denoising methods on target-domain noise data.

Index Terms—Convolutional neural network (CNN), noise attenuation, seismic exploration, target-domain noise, transfer learning.

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Xintong Dong, Ming Cheng, Hongzhou Wang, and Jun Lin are with the College of Instrumentation and Electrical Engineering, Jilin University, Changchun 130026, China, and also with the Southern Marine Science and Engineering Guangdong Laboratory (Zhanjiang), Zhanjiang 524000, China (e-mail: dxt@jlu.edu.cn; chengming22@mails.jlu.edu.cn; hzwang21@mails.jlu.edu.cn; lin_jun@jlu.edu.cn).

Guanghui Li is with the College of Physics and Electronic Engineering, Shanxi University, Taiyuan 030006, China (e-mail: ligh1986@sxu.edu.cn).

Shaoping Lu is with the School of Earth Sciences and Engineering and the Guangdong Provincial Key Laboratory of Geodynamics and Geohazards, Sun Yat-sen University, Guangzhou 510275, China, and also with the Southern Marine Science and Engineering Guangdong Laboratory (Zhuhai), Zhuhai 519000, China (e-mail: lushaoping@mail.sysu.edu.cn).

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I. INTRODUCTION

SEISMIC background noise arises from numerous sources, ranging from acquisition environment to human activities to undesirable waves excited by the source (e.g., ground rolls and multiples) [1], [2], [3], [4]. Background noise is ever presented in seismic data and seriously contaminates the desired signals utilized to obtain the information of subsurface structure [1].

The community of exploration geophysics has developed a number of denoising methods for seismic data over the years; these methods can significantly enhance the signal-to-noise ratio (SNR) of seismic data and then provide reliable data for the following high-resolution seismic imaging [5], [45]. Representative seismic denoising approaches include f-k predictive filtering [6], f-x deconvolution [7], empirical mode decomposition (EMD) [8], [9], singular value decomposition (SVD) [10], analytical transform [11], [12], [13], dictionary learning [14], and some rank-reduction methods (singular spectrum analysis, Cadzow filtering, principal component analysis, and so on) [15], [16], [37], [38]. Although these proposed seismic denoising methods have exhibited outstanding performance, some limitations still hinder their further applications in handling some complex seismic data with low SNR. For instance, unsuitable filter length is likely to cause the performance degradation of f-k predictive filtering [17]. Discarding some decomposed modes may lead to the decay of recovered signals while utilizing EMD-based denoising methods; this phenomenon may occur with high noise level [2]. Analytical transforms (e.g., wavelet, curvelet, contourlet, dreamlet, and shearlet) are heavily dependent on pursuing the optimal threshold function, but it is challenging to obtain relevant prior knowledge before application [12]. Dictionary learning adaptively learns the basis from the processed seismic data itself instead of adopting the fixed basis, thus showing superior denoising performance to the analytical transforms. However, thousands of SVD operations required by dictionary updating dramatically aggravate its computational expense [14]. Furthermore, most of these existing denoising methods are based on some certain assumptions, so artificial parameter adjustments are required to generate the best possible denoised result [5]. For example, the essential precondition of f-x deconvolution is that the event is linear or local linear [10]. Rank-reduction methods assume that the seismic record without noise is an approximately low-rank structure, while

noise contamination will increase its rank [15]. The basis of analytical transform and dictionary learning is the sparsity of seismic data [14]. Obviously, a large number of real complex seismic records do not satisfy these aforementioned strict assumptions; moreover, the artificial parameter adjustment is time-consuming and depends on human experience, which is not conducive to denoise the high-dimensional and large-scale seismic data.

Following the great success in numerous domains of data processing, deep learning (DL) denoising methods for seismic data have attracted a lot of attention from the community of geophysics [44]. Such methods adopt sufficient and high-quality training data to optimize the trainable parameters of convolutional neural network (CNN), thereby achieving the intelligent denoising of seismic data without parameter fine-tuning via the trained model [10], [18], [19]. Dong et al. [2] developed an adaptive denoising CNN based on the determination of high-order statistics to attenuate the low-frequency noise in desert seismic data acquired from the Tarim basin. Saad and Chen [20] proposed an alternative for the suppression of seismic random noise based on the deep denoising autoencoder. Kaur et al. [3] and Li et al. [21] utilized the cycle generative adversarial network (GAN) with unpaired training data to attenuate the surface waves. Liu et al. [19] proposed a promising unsupervised DL denoising paradigm without using noise-free data, showing impressive performance in both prestack and poststack seismic data. Cheng et al. [22] designed a multiscale denoising network, which can extract features from seismic data with different resolutions and then apply this novel network architecture to denoise the seismic data received by the distributed acoustic sensor. Compared with the aforementioned traditional methods, in most cases, DL-based denoising methods have shown better performance in noise attenuation, weak signal recovery, SNR improvement, and automation (e.g., without artificial parameter fine-tuning) [19], [22]. Generally, the training sample is one major component in these DL-based denoising methods, so huge distribution difference between training and testing datasets will significantly degrade their performance [5], [23]. Therefore, we need to prepare sufficient noise training data whose distribution is the same as the noise appearing in the testing seismic data before using these DL denoising methods, especially the supervised fashion. However, such noise training data with the same distribution, called target-domain noise data in this article, is often unobtainable in the real condition mainly due to the three reasons: 1) real seismic data often contain numerous kinds of noise owing to different sources [1]; 2) the same kind of seismic noise acquired from different environments exhibits differences in some characteristics, including dominant frequency, power spectral distribution, Gaussianity, and linearity [24]; and 3) confidentiality considerations. To the best of our knowledge, previous studies mainly adopt three approaches to generate the noise dataset used for training. First, white Gaussian noise (WGN) is sometimes utilized to act as the target-domain noise data [20]; obviously, this approach will seriously degrade the performance of DL-based methods when the noise in testing seismic data does not conform to the Gaussian distribution. Another alternative,

some experts have adopted, is to filter some noise data from the testing seismic data via a conventional denoising method, but the denoising performance of trained model is, at best, equal to the conventional denoising method used to generate the noise training data [25]. Third, the best choice is the passive source data matched with the testing seismic data [5], [21]; such data are basically equivalent to the target-domain noise data, but often unavailable in real seismic exploration. More recently, some experts exert a great deal of effort in utilizing unsupervised or self-supervised paradigm to alleviate this problem: lack of target-domain noise data [21], [25]. Some studies have proven the feasibility of these methods and they usually outperform conventional seismic denoising methods, but their robustness and denoising performance are often slightly inferior to the supervised fashion when we have enough high-quality training data. Furthermore, supervision is a relatively more accessible training flow compared with unsupervision and self-supervision. On the whole, it is necessary to explore the solution for the limited target-domain noise data within the framework of supervised DL.

When utilizing DL-based methods to complete some tasks including classification, regression, and clustering, we often face a challenge: the training and testing data have different distributions, especially when facing some real-world applications [26], [27]. In other words, we sometimes focus on a task in one domain of interest, but we only have sufficient training data in another domain. Transfer learning attempts to transfer the features learned from some source domains to a target domain of interest when the latter do not have sufficient high-quality training data [26]. Apparently, transfer learning is a potential approach that can solve the problem mentioned above: lack of target-domain noise data. However, some relevant studies also demonstrate the weak correlation (i.e., huge distribution difference) between source and target domains will dramatically affect the performance of transfer learning. Hence, the prerequisite for its successful applications in seismic data denoising is that we have sufficient noise training data in the source domain and its distribution is relatively similar to that of target-domain noise data.

Inspired by the above strategy, we propose a novel DL paradigm for seismic data denoising based on transfer learning and noise modeling method [28]; the latter can provide sufficient synthetic noise data for the source domain. Specifically, we first design a multicascade residual attention network (MCRA-Net); this new architecture adopts the downsampling operation to generate three inputs with different resolutions, and then, convolution operations are applied to capture rich multiresolution features [30], [33] from these three inputs. Second, the Green-function-based noise modeling method proposed by our group [28] is used to simulate a large number of seismic noise records; we utilize these synthetic noise records and numerous theoretical noise-free records obtained by forward modeling to train the network and thus obtain the pretrained model in source domain. Finally, a small amount of target-domain noise data extracted from a passive source record is utilized to fine-tune the partial trainable parameters of pretrained model, so as to transfer the pretrained model from source domain to target domain. In this article, we call

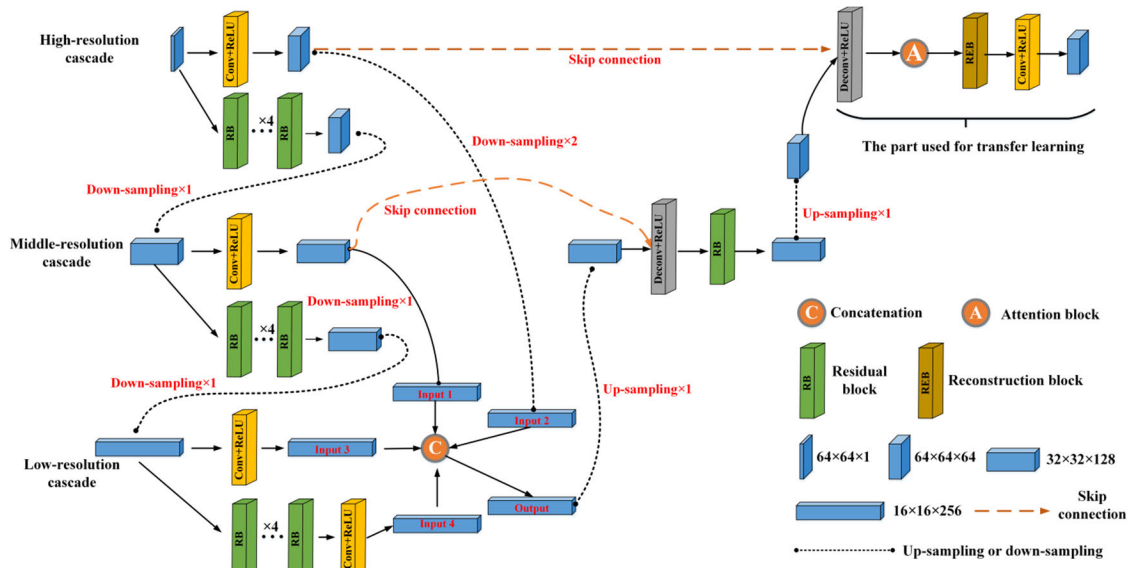


Fig. 1. Architecture of MCRA-Net.

this novel DL paradigm as transfer learning driven by noise modeling (TL-NM). In both synthetic and real examples, this proposed TL-NM exhibits almost the same denoising result as leveraging a large amount of target-domain noise data to train MCRA-Net and significantly better performance than conventional seismic denoising methods. Furthermore, the new network architecture, MCRA-Net, shows some advantages over two existing architectures of CNN.

II. NETWORK ARCHITECTURE

It is known that extracting more beneficial features is important for the performance and efficiency of CNN-based methods [29]. Inspired by this view, as shown in Fig. 1, this proposed MCRA-Net adopts downsampling operations to generate feature maps with different resolutions: 64×64 , 32×32 , and 16×16 . Then, a number of Conv+ReLU layers and residual blocks (RBs) are performed on these multiresolution feature maps to extract more useful features compared with just considering a single resolution; we call these extracted features as multiresolution features in this article. Obviously, high- and low-resolution features correspond to the local fine and global coarse information of seismic data, respectively. Next, we focus on five major components of the MCRA-Net: multicascade structure, RB, attention block, reconstruction block (REB), and loss function.

A. Multicascade Structure

The multicascade structure is an important basis for obtaining the conducive multiresolution features [30], [31]. Specifically, at high-resolution cascade, one Conv+ReLU layer and four RBs in series are first utilized to capture high-resolution features from the input data with size of $64 \times 64 \times 1$; the two outputs of high-resolution cascade are with the size of $64 \times 64 \times 64$. Afterward, we perform one downsampling operation on the output of four RBs in high-resolution cascade to generate the input data of middle-resolution cascade with

a size of $32 \times 32 \times 128$. It is noticed that we increase the number of channels from 64 to 128 to mitigate the possible information loss caused by size shrinkage. Similarly, the input data of low-resolution cascade are generated by applying one downsampling operation to the output of four RBs in middle-resolution cascade, thereby extracting global coarse features from the low-resolution input data with a size of $16 \times 16 \times 256$. Finally, we leverage a concatenation operation to fuse the two outputs of low-resolution cascade (i.e., inputs 3 and 4 in Fig. 1), the output of Conv+ReLU layer in high-resolution cascade after being applied two downsampling operations (i.e., input 2 in Fig. 1), and the output of Conv+ReLU layer in middle-resolution cascade after being applied one downsampling operation (i.e., input 1 in Fig. 1). On the whole, this multicascade structure enables the proposed MCRA-Net to capture a large number of favorable multiresolution features used for the separation of signals and noise.

B. Attention Block

Complex background noise makes it extremely difficult to extract suitable features for the task of seismic data denoising [5]. Hence, we deploy a lightweight attention block combined with depthwise separable convolution [32] to strengthen the expression ability of MCRA-Net. The main function of this attention block is to highlight the features conducive to the denoising task and also discard the redundant features. The concrete structure of attention block is shown in Fig. 2(a). In this attention block, we adopt the depthwise separable convolution to map the output of Conv+ReLU layer with a size of $64 \times 64 \times 64$ into four 3-D spaces. Then, these four separated spaces with a size of $64 \times 64 \times 16$ are fed into four parallel Conv+ReLU layers; the corresponding four outputs are fused by using a concatenation operation. Finally, the softmax function generates a probability distribution matrix used to multiply the input data of attention block, so as

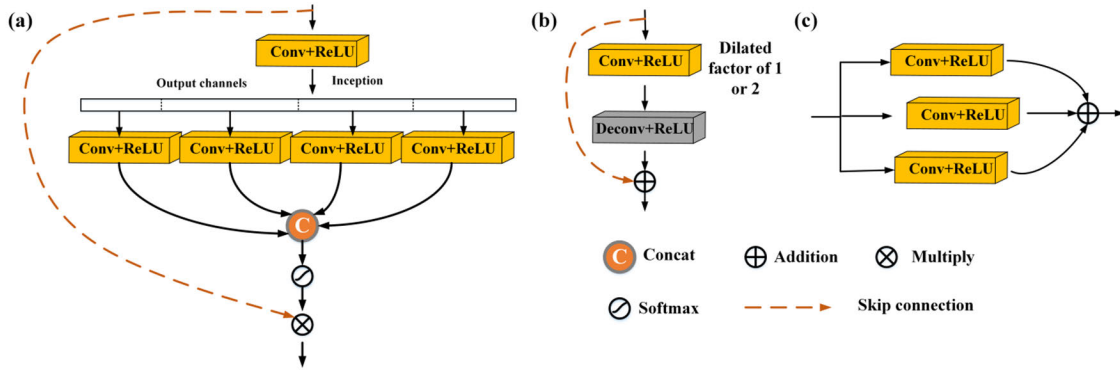


Fig. 2. (a)–(c) Structures of attention block, RB, and REB, respectively.

to allocate larger weights on the desired features. The main function of depthwise separable convolution is to alleviate the extra computational cost caused by the attention mechanism almost without sacrificing the precision of trained model [32].

C. Residual Block

In each cascade, four continuously connected RBs are used to capture multiresolution features; the concrete structure of RB is shown in Fig. 2(b). Enlarging the receptive field is a commonly used approach to capture more context information in numerous CNN-based methods, thus further reinforcing their performance [33]. Therefore, we deploy some dilated convolutions in these RBs. Specifically, in each group of RBs, the first and third adopt the standard convolution of 3×3 and dilated factor of 1, while the second and fourth are with a convolution of 3×3 and a dilated factor of 2. In addition, some skip connections are utilized in these RBs to avoid the overfitting phenomenon so that we can enhance the denoising performance of MCRA-Net by stacking more layers.

D. Reconstruction Block

As shown in Fig. 2(c), the REB of MCRA-Net is a concise three-layer parallel Conv+ReLU structure that functions on the high-resolution feature map with a size of $64 \times 64 \times 64$. We utilize this simple and effective architecture to reconstruct the desired signals.

E. Loss Function

As shown in the following equation, the raw seismic data y can be expressed as the sum of signals x and noise n :

$$y = x + n. \quad (1)$$

In this study, we utilize the paired training data $\{P_x^i, P_n^i\}_{i=1}^N$ and mean square error (mse) to generate the trained model in a supervised fashion, where P_x^i and P_n^i represent the i th given signal and noise training patches, respectively. The implementation of training process can be formulated as [22], [34]

$$L_{\text{MSE}}(\theta) = \frac{1}{2N} \sum_{i=1}^N \|f_{\text{MCSR-Net}}(P_x^i + P_n^i) - P_x^i\|_F^2 \quad (2)$$

where θ represents the trainable parameters of MCRA-Net and $\|\cdot\|_F$ denotes the Frobenius norm.

III. STRATEGY OF TRANSFER LEARNING

The research on transfer learning is motivated by the fact that human can intelligently apply previously learned knowledge to solve new problems [26]. Many tasks of natural image processing, such as denoising, super-resolution, and fusion, have demonstrated two viewpoints: 1) transfer learning allows the domains or distributions used in training and testing to be different and 2) although transfer learning can help CNN obtain the optimal trained model without using a large amount of training data in the target domain, the huge distribution difference between the source domain and the target domain will still decrease its performance. Based on these two viewpoints, we design a novel transfer-learning-based training workflow for MCRA-Net from two aspects. On the one hand, transfer learning strategy is used to transfer synthetic seismic noise from source domain to target domain. On the other hand, before applying the transfer learning, we need to generate a massive amount of synthetic seismic noise whose distribution is relatively similar to that of the noise presented in testing seismic data (i.e., the target-domain noise data mentioned above). Specifically, we first utilize a seismic noise modeling method based on the Green function [28] to simulate a large number of synthetic seismic noise records close to the real condition. Then, these synthetic noise records are used for the pretraining in source domain, so as to generate the pretrained model via the mse loss function. Finally, some limited target-domain noise is used to transfer the pretrained model from source domain to target domain by performing fine-tuning operation on it. We call this novel training workflow as TL-NM. The general strategy of TL-NM is shown in Fig. 3 and its detailed information will be described next.

A. Pretraining in Source Domain

1) *Signal Dataset*: As shown in (2), paired signal and noise patches are used to optimize the trainable parameters of MCRA-Net in a supervised fashion. For the signal dataset, we utilize the forward modeling method to generate theoretical clean seismic records. First, to ensure the authenticity and representation of signal dataset, we construct 70 forward models with different subsurface structures (flat stratum, fault, anticline, syncline, and so on) and P-wave velocity distributions; the P-wave velocity ranges from 1300 to 5200 m/s. Then, three kinds of seismic wavelet, Ricker, single, and symmetrical

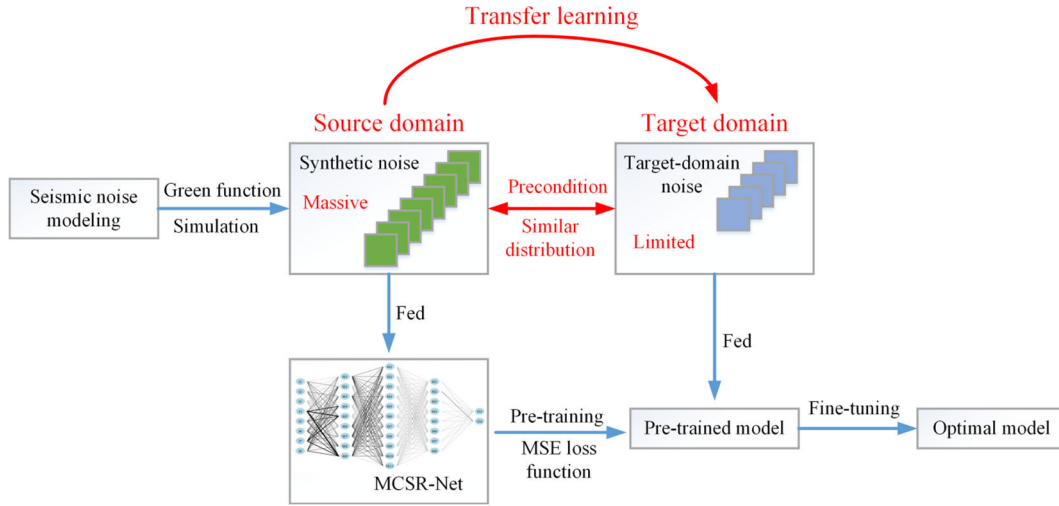


Fig. 3. General strategy of TL-NM.

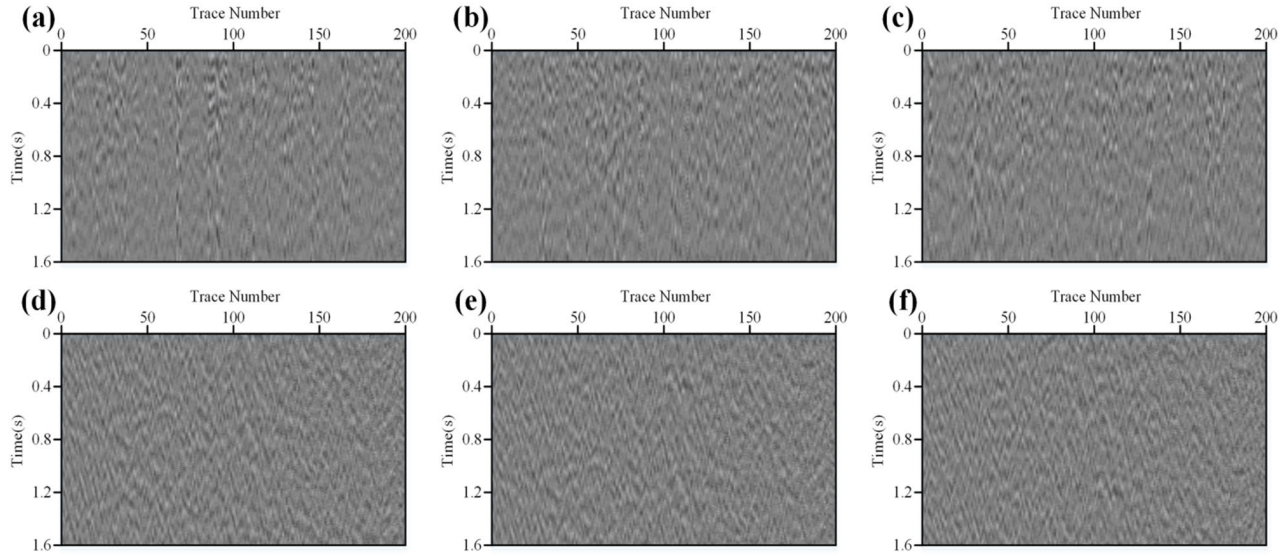


Fig. 4. (a)–(c) Three synthetic random noise records generated by using noise modeling method based on the Green function. (d)–(f) Three real random noise records extracted from the adopted passive source record. Intuitively, synthetic noise is similar to real noise.

wavelets with dominant frequency ranging from 15 to 35 Hz, are used to activate these aforementioned forward models, thus generating numerous theoretical clean seismic records by solving the acoustic wave equation; the sampling rate and trace interval of these clean records are 2 ms and 10, 20, or 30 m, respectively. Finally, we extract 25 000 signal patches with a size of 64×64 pixels from these synthetic clean records; these extracted signal patches after being applied normalized operation are the signal dataset of MCRA-Net.

2) *Synthetic Noise Dataset Generated by Noise Modeling:* In the later experimental part, we mainly use random noise to testify the effectiveness of TL-NM. Therefore, a modeling method for seismic random noise proposed by our group [28] is utilized to generate synthetic noise data close to the real condition. According to the Green function, the source of seismic random noise can be assumed as numerous point-like sources distributed around the receiver [35]. Based on this assumption, a seismic random noise record is taken as the superimposed

wave field excited by a number of independent sources in a homogeneous isotropy half-infinite surface. Hence, we can obtain a number of synthetic seismic random noise records by solving the inhomogeneous wave equation. More detailed information of this random noise modeling method is available in the Appendix and [28]. In this article, we classify the seismic random noise into three categories: wind noise, near-field noise, and far-field noise; concrete parameter settings of this random noise modeling are shown in Table I. First, we adopt this random noise modeling method based on the Green function to generate 80 synthetic random noise records with a time length of 1.6 s and a number of traces 200; three of them are shown in Fig. 4(a)–(c). Second, 25 000 synthetic noise patches with a size of 64×64 pixels are intercepted from these 80 noise records. Third, the 25 000 synthetic noise patches and 25 000 signal patches are fed to the proposed MCRA-Net, so as to obtain the pretrained model after 50 epochs; the corresponding mse loss curve is shown

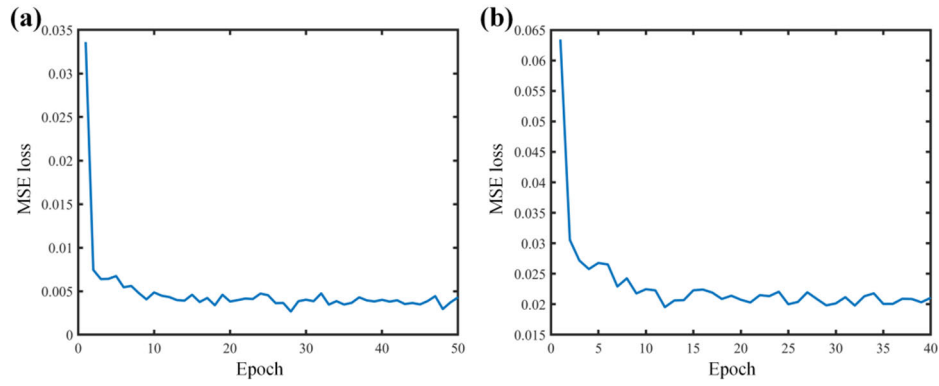


Fig. 5. (a) and (b) MSE loss curves of pretraining and fine-tuning, respectively.

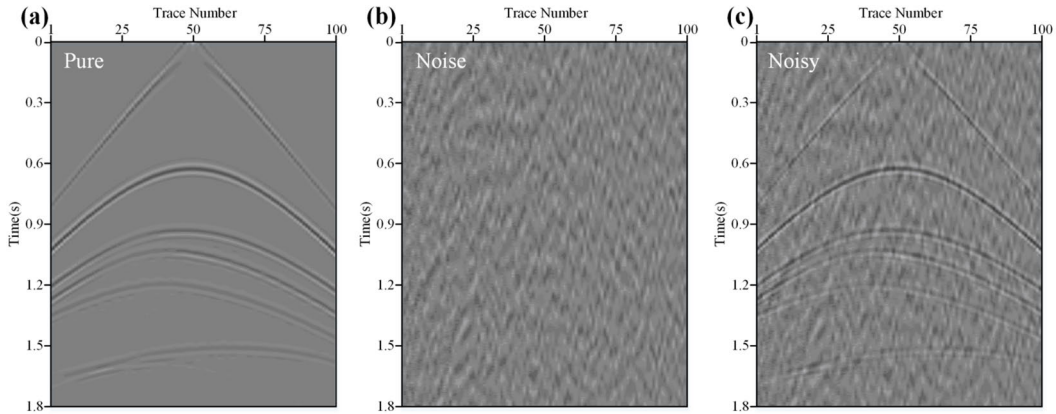


Fig. 6. Synthetic example. (a) Synthetic theoretical clean record. (b) Real random noise record. (c) Synthetic noisy seismic record.

TABLE I
KEY PARAMETERS OF RANDOM NOISE MODELING

Noise type	Parameters	Specifications
near-field noise	central frequency(Hz)	1-20
	source distribution radius (m)	200-1250
far-field noise	central frequency(Hz)	1-200
	source distribution area (m)	4000×750
wind noise	roughness coefficient	0.1-0.15
	drag-force coefficient	0.01-0.018
	wind velocity (m/s)	300-600

in Fig. 5(a). Concrete hyperparameters of the pretraining are given as follows: network depth 36, size of convolutional kernel 3×3 , dilated factor 2, patch size 64×64 , batch size 128, and learning rate $[10^{-3}, 10^{-5}]$.

B. Fine-Tuning in Target Domain

In this section, some real seismic random noise is used to transfer the pretrained model from source domain to target domain. Passive source record is acquired under the condition of no source; it mainly contains the interference of surrounding environment and thus can be approximated as random noise in the seismic record. Thus, we utilize a passive source record as the target-domain noise data used for fine-tuning; three real random noise records extracted from this passive source record are shown in Fig. 4(d)–(f). We randomly extract 1500 real random noise patches from this passive source record by

using a sliding window with a size of 64×64 pixels. These real random noise patches and signals patches are utilized to fine-tune the trainable parameters of the part used for transfer learning (see Fig. 1), so as to obtain the optimal model in the target domain after 40 epochs; the mse loss curve is shown in Fig. 5(b). In this article, both the training and testing experiments of TL-NM are conducted in PyTorch 1.10.0 with CUDA 11.0 running on a PC with 11th Gen Intel¹ Core i7-11800H CPU and NVIDIA RTX 3080 GPU.

IV. SYNTHETIC EXAMPLE

We leverage a Ricker wavelet with a dominant frequency of 25 Hz to activate a forward model whose P-wave velocity ranges from 1200 to 4200 m/s. Fig. 6(a) shows the corresponding theoretical clean seismic record generated by solving the acoustic wave equation; its sampling rate, trace interval, and time length are 2 ms, 20 m, and 1.8 s, respectively. Fig. 6(b) shows that a real random noise record was also extracted from the passive source record mentioned above; it is noticed that this noise record does not contain the 1500 real random noise patches used for the fine-tuning in target domain, so as to guarantee the independence between training and testing. The synthetic noisy seismic record shown in Fig. 6(c) is the superposition of Fig. 6(a) and (b); its SNR and root

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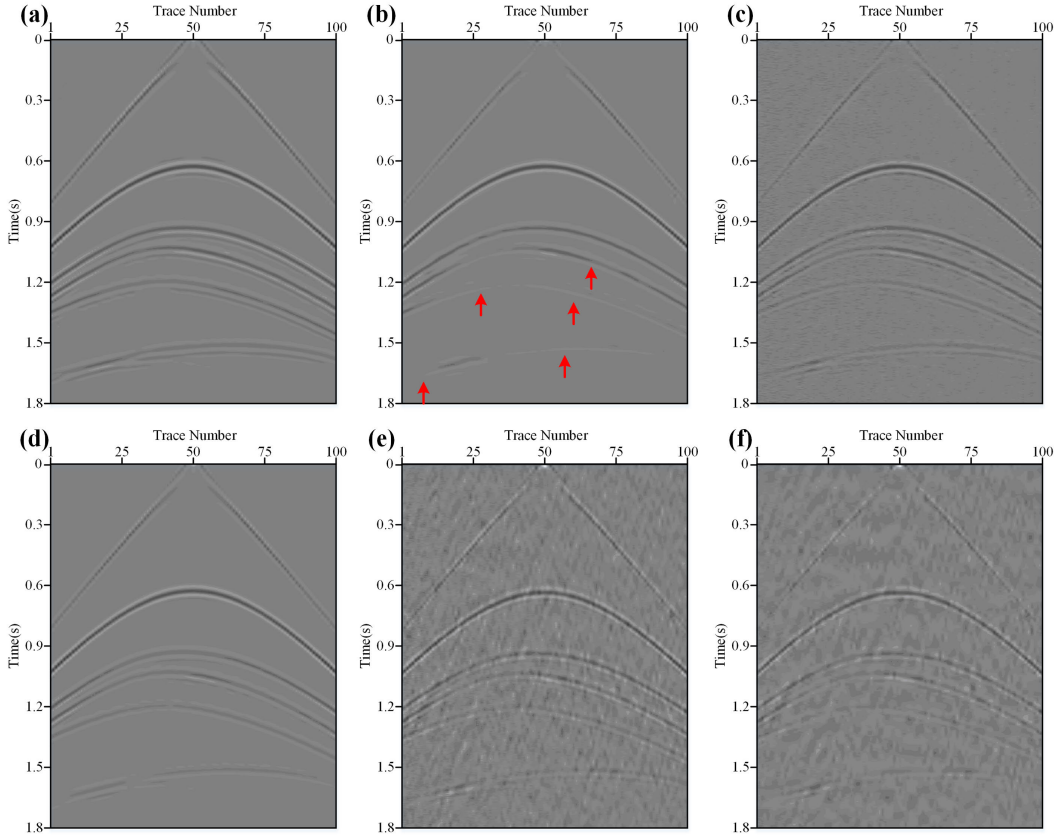


Fig. 7. Comparison of synthetic denoised results. (a)–(f) Denoised results of Fig. 6(c) after applying TL-NM (ours), pretrained model, WGN model, target-domain noise model, RPCA, and wavelet transform, respectively.

mse (RMSE) are -5 dB and 0.2107, respectively. These two measurements can be formulated as [2], [10]

$$\text{SNR(dB)} = 10 \lg \left(\frac{\sum_{i=1}^M x^2(i)}{\sum_{i=1}^M [d(i) - x(i)]^2} \right) \quad (3)$$

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^M [d(i) - x(i)]^2}{M}} \quad (4)$$

where $x(i)$ stands for clean seismic record, $d(i)$ represents noisy record or denoised result, and M is the number of sampling points. Larger SNR and smaller RMSE indicate better denoising performance.

A. Competitive Methods

In this section, the proposed TL-NM is compared with the pretrained model to demonstrate the effect of transfer learning. In addition, we compare TL-NM with two trained models also derived from the proposed MCRA-Net to illustrate the importance of noise dataset. These two trained models adopt different noise datasets and their other settings (signal dataset, hyperparameter, loss function, and so on) are the same as TL-NM. For one of them, we utilize 25 000 64×64 pixels WGN patches to feed the MCRA-Net; this trained model is named WGN model. For the other one, we use a large amount of target-domain noise data to act as the noise dataset. Specifically, 25 000 64×64 pixels real seismic random noise patches extracted from the passive source record are fed to

the MCRA-Net and the trained model is called target-domain noise model in this article.

Furthermore, we also compare TL-NM with two conventional seismic denoising methods: robust principal component analysis (RPCA) [39], [40] and wavelet transform [41]. RPCA is a classical rank-reduction method, which can decompose noisy seismic record into a low-rank matrix and a sparse matrix, so as to obtain the denoised result by retaining the low-rank matrix and discarding the sparse matrix; the weight on the sparse error term of RPCA is 0.025 in this synthetic example. Two important settings of wavelet transform are basic function db4 and adaptive soft threshold.

B. Comparison of Denoised Results

We utilize TL-NM, pretrained model, WGN model, target-domain noise model, RPCA, and wavelet transform to handle the noisy record shown in Fig. 6(c). The six denoised results are shown in Fig. 7(a)–(f) successively. As shown in Fig. 7(a) and (d), TL-NM and target-domain noise model exhibit similar denoising performance; the two can suppress most of the random noise and also recover the signals completely, so as to achieve a good tradeoff between noise suppression and signal amplitude preservation. This similarity demonstrates that transfer learning can help CNN to generate the optimal trained model without using a large amount of target-domain noise data. In Fig. 7(b), the pretrained model has good performance in attenuating the random noise, but some weak

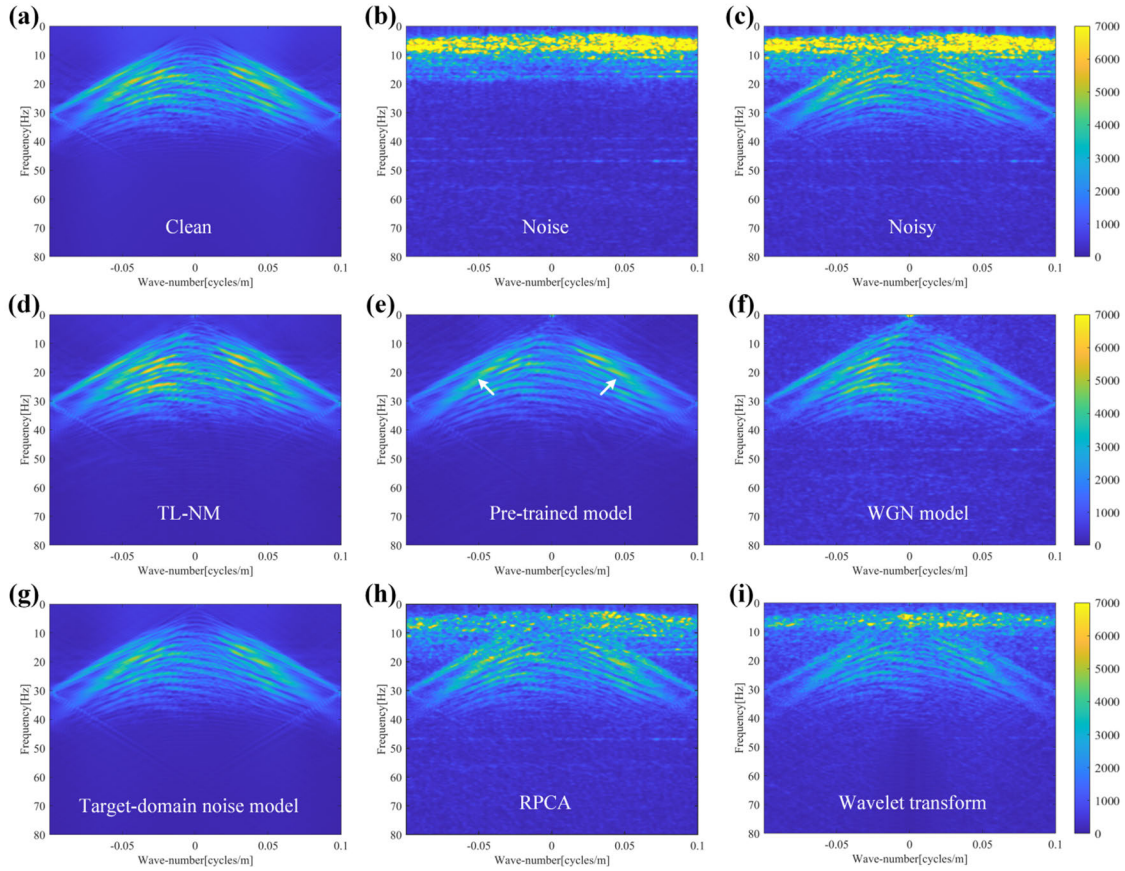


Fig. 8. Frequency-domain analysis. (a) F–K spectrum of the theoretical clean record in Fig. 6(a). (b) F–K spectrum of the noise record in Fig. 6(b). (c) F–K spectrum of the noisy record in Fig. 6(c). (d)–(i) F–K spectrum of the six denoised results shown in Fig. 7(a)–(f), respectively.

TABLE II
COMPARISON OF SNRS AND RMSE

Method	TL-NM	Pre-trained	WGN	Target-domain	RPCA	Wavelet
SNR	12.1338	8.4948	5.3660	12.6399	-0.1487	0.5783
RMSE	0.0293	0.0293	0.0639	0.0277	0.1206	0.1109

signals indicated by red arrows are not recovered well and the continuity of events needs to be enhanced further. This performance degradation illustrates that although the synthetic random noise generated by the Green-function-based noise modeling method is similar to the real condition, it still cannot completely replace the target-domain noise data in network training. Lots of residual random noise in Fig. 7(c) shows that the added noise in Fig. 6(b) does not conform to the Gaussian distribution. Obviously, the huge distribution difference between training and testing datasets will dramatically degrade the denoising performance of CNN-based methods. As shown in Fig. 7(e) and (f), these two conventional methods only remove partial random noise and it is still difficult to identify the reflected signals.

To further quantify the denoising performance of six methods, we display the SNR and RMSE of the six synthetic denoised results [see Fig. 7(a)–(f)] in Table II. SNR and RMSE of the denoised result by using TL-NM are extremely close to those of the denoised result by using the target-domain noise model, indicating their similar denoising performance

and the potential of TL-NM in alleviating the dependence on target-domain noise data. Furthermore, TL-NM exhibits higher SNR and smaller RMSE compared with the four competitive methods (i.e., WGN model, pretrained model, RPCA, and wavelet transform), demonstrating its best denoising performance among these five methods.

Next, we analyze the denoising performance of different methods in the frequency domain. Fig. 8(a)–(c) shows the F–K spectrum of the clean record in Fig. 6(a), the real noise record in Fig. 6(b), and the noisy record in Fig. 6(c), respectively. We can discover from Fig. 8(c) that signals and noise coexist in the low-frequency band ranging from 0 to 20 Hz and it is challenging to achieve complete separation of the two in shared frequency band. Fig. 8(d)–(i) shows the F–K spectrum of the six denoised results shown in Fig. 7(a)–(f), respectively. The F–K spectrum shown in Fig. 8(d) and (g) is extremely similar to the F–K spectrum shown in Fig. 8(a), demonstrating that TL-NM and target-domain noise model effectively separate signals from noise in the shared frequency band. As shown in Fig. 8(e), we can still observe a small amount of residual noise in the F–K spectrum of denoised result by using a pretrained model. Moreover, as indicated by white arrows, the energy of signals is attenuated after denoising. Fig. 8(f) once again proves the residual noise in the denoised result of WGN model. In Fig. 8(h) and (i), a massive amount of random noise still seriously contaminates the signals, especially in the 0–20-Hz low-frequency band.

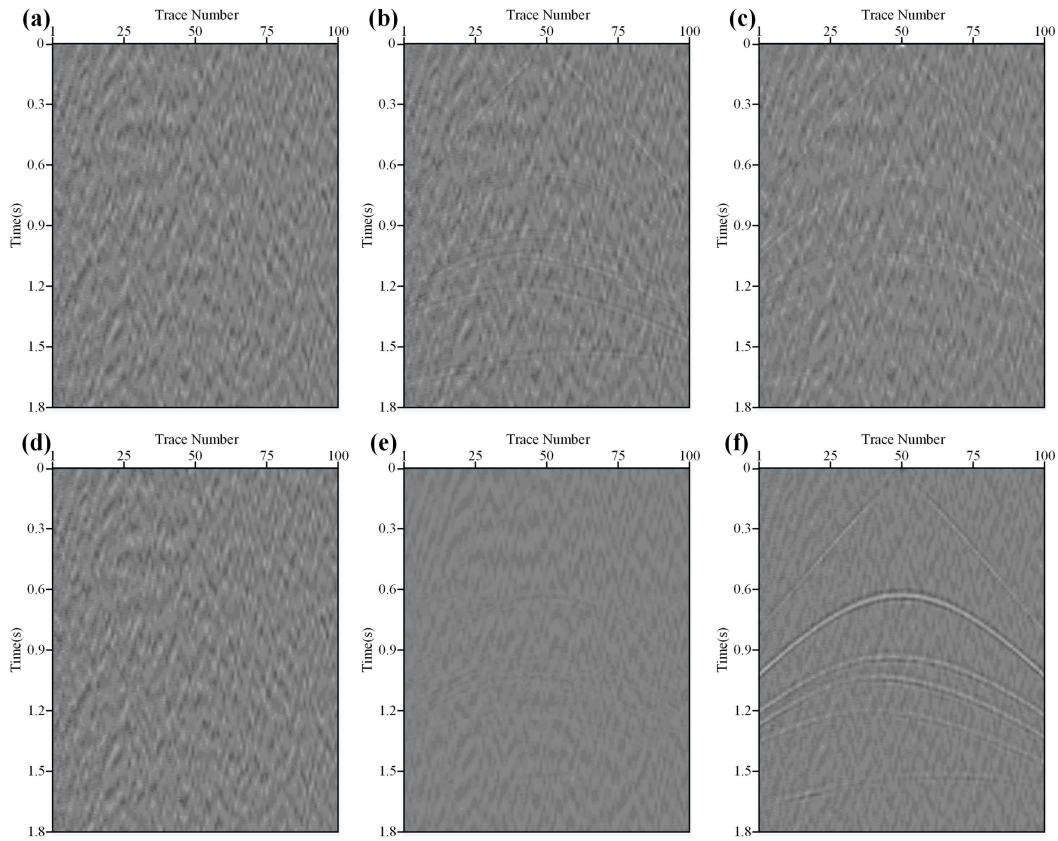


Fig. 9. Comparison of difference records. (a)–(f) Difference records corresponding to TL-NM, pretrained model, WGN model, target-domain noise model, RPCA, and wavelet transform, respectively.

To compare the signal preservation ability of different methods, we display the difference records in Fig. 9 [Fig. 6(c) minus Fig. 7(a)–(f), respectively]. Residual signals in Fig. 9(b), (c), (e), and (f) illustrate that the pretrained model, WGN model, RPCA, and wavelet transform lead to the decay of signal amplitude when attenuating the random noise. On the contrary, we barely observe any signals in the two difference records shown in Fig. 9(a) and (d), and these two difference records are almost equal to pure noise record, which suggests that both TL-NM and target-domain noise model achieve a good tradeoff between noise suppression and signal preservation.

V. REAL EXAMPLE

In this section, we utilize a real 3-D shot gather to testify the effectiveness of TL-NM. It is noted that the passive source record used for the fine-tuning of pretrained model and the adopted shot gather are acquired from the same surveying area of northwestern China, that is, random noise patches intercepted from this passive source record are the target-domain noise data aiming at this shot gather used for testing. Furthermore, this passive source record and 3-D shot gather are acquired from a desert area of China. Loose sand layer on the surface can absorb some high-frequency components, resulting in the energy of random noise being mainly distributed in the low-frequency band. This phenomenon has been demonstrated

by the F–K spectrum shown in Fig. 8(b). Fig. 10(a) shows a real common-shot-point (CSP) record extracted from the 3-D shot gather; its trace interval and sampling rate are 20 m and 2 ms, respectively. In this real seismic record, random noise mainly caused by wind and surface waves simultaneously destroys the reflected signals, leading relatively low SNR of the whole record. We use TL-NM, pretrained model, WGN model, target-domain noise model, RPCA, and wavelet transform to process this CSP record, and their denoised results are shown in Fig. 10(b)–(g), respectively. The four DL denoising methods adopt the original trained models used in the synthetic example and without extra training; the weight on the sparse error term of RPCA is adjusted to 0.028. In Fig. 10(b) and (e), both TL-NM and target-domain noise model can remove almost all random noise and surface waves; the background of these two denoised results becomes clear and the continuity of events is greatly strengthened. It can be concluded that the denoising performance of TL-NM is similar to that of using a massive amount of target-domain noise data. In the denoised result after applying the pretrained model [see Fig. 10(c)], residual random noise still interferes with signals and the background is not clear enough; some weak events in red boxes are still difficult to observe. In Fig. 10(d), obvious residual prearrival noise in red boxes demonstrates the poor performance of WGN model in random noise suppression. Moreover, residual surface waves seriously contaminate the reflected signals and some events indicated by red arrows

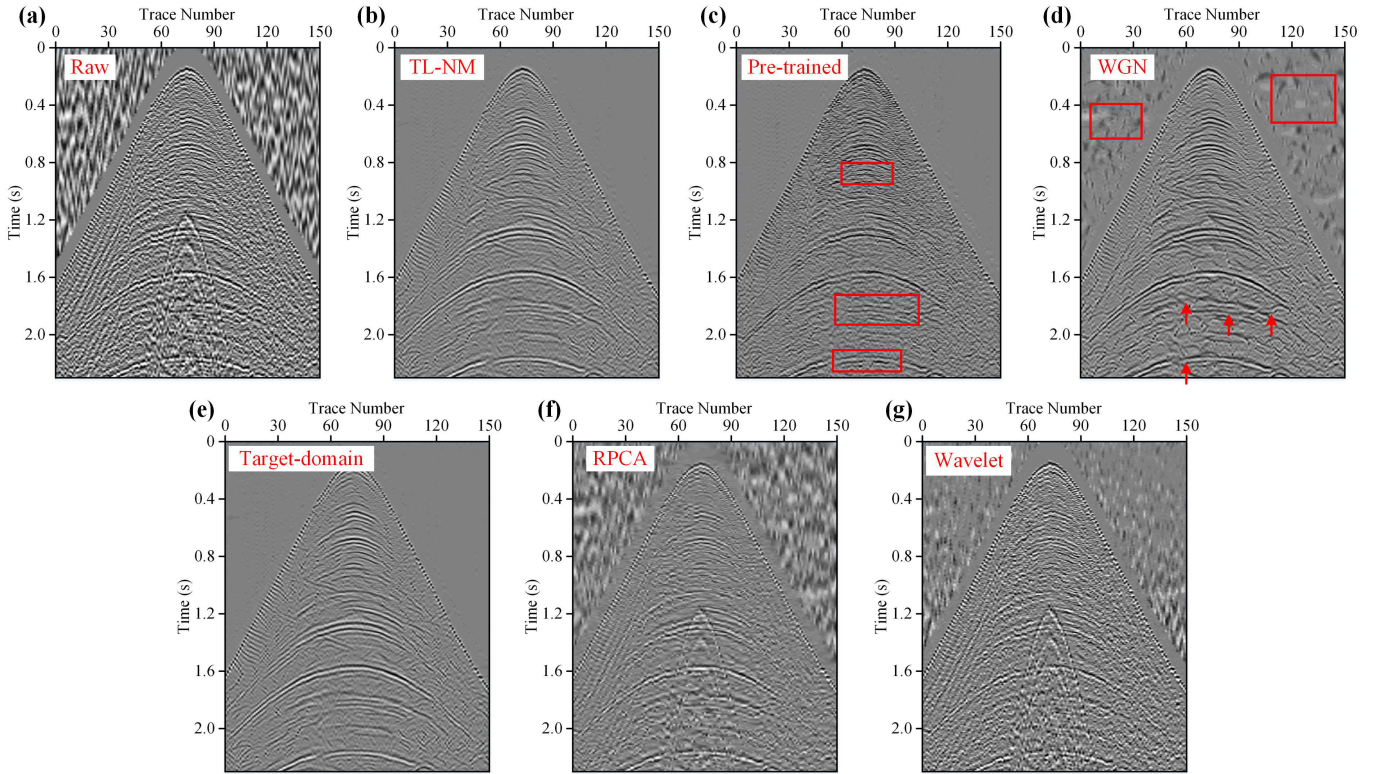


Fig. 10. Denoised results of the real CSP record after applying different methods. (a) Real CSP record. (b)–(g) Denoised results of Fig. 10(a) after applying TL-NM, pretrained model, WGN model, target-domain noise model, RPCA, and wavelet transform, respectively.

exhibit poor continuity. Obviously, as shown in Fig. 10(e) and (f), the denoised results of the two conventional methods are significantly worse than those of the four DL methods. A large amount of background noise still remains in these two denoised results after applying RPCA and wavelet transform.

In Fig. 10(b) and (e), some tiny crossed events appear in the denoised results by using TL-NM and target-domain noise model. By comparing the two denoised results with the raw CSP record in Fig. 10(a), we deem that the main cause of these tiny crossed events is some tiny reflectors in stratum, instead of denoising operation.

Fig. 11 shows the difference records obtained by subtracting the six denoised results [see Fig. 10(b)–(e)] from the CSP record [see Fig. 10(a)]. In Fig. 11(b), (c), (e), and (f), signal leakages indicated by red arrows demonstrate that pretrained model, WGN model, RPCA, and wavelet transform attenuate the amplitude of signals. On the contrary, the phenomenon of signal leakage in Fig. 11(a) and (d) is obviously alleviated, demonstrating the stronger ability of TL-NM and target-domain noise model to signal amplitude preservation.

Next, we extract another CSP record shown in Fig. 12(a) from the 3-D shot gather to verify the generalization of TL-NM. Denoised results after applying the six methods are shown in Fig. 12(b)–(g) and the corresponding difference records are shown in Fig. 13. As shown in Figs. 12(b) and (e) and 13(a) and (d), TL-NM and target-domain noise model still exhibit good denoising performance: complete noise suppression, continuous recovered events, and almost no signal leakage in the difference records. The pretrained model and the

WGN model are still disturbed by the distribution difference between training and testing datasets, causing some residual noise in Fig. 12(c) and (d) and observable signal leakages in Fig. 13(b) and (c). Obvious residual noise in Fig. 12(f) and (g) and signal leakages in Fig. 13(e) and (f) once again demonstrate the worse denoising performance of RPCA and wavelet transform compared with the other four CNN-based methods.

VI. DISCUSSION

A. Importance of Green-Function-Based Noise Modeling Method

The motivation of transfer learning is to extend the application scope of machine learning when distributions of training and testing datasets are different. However, it is still difficult to maintain the good performance of transfer learning when suffering from obvious distribution differences. To testify to this viewpoint, we perform transfer learning operation on the WGN model. Specifically, we leverage 2000 64×64 pixels target-domain noise patches extracted from the passive source record to fine-tune the trainable parameters of WGN model, thus obtaining the trained model (called model 1). Model 1 is used to process the two real CSP records shown in Figs. 10(a) and 12(a); the corresponding two denoised results are shown in Fig. 14. Residual background noise (random noise and surface waves) and discontinuous events in Fig. 14(a) and (b) suggest that the performance of model 1 is inferior to that of TL-NM. Obviously, this performance degradation is caused by the huge

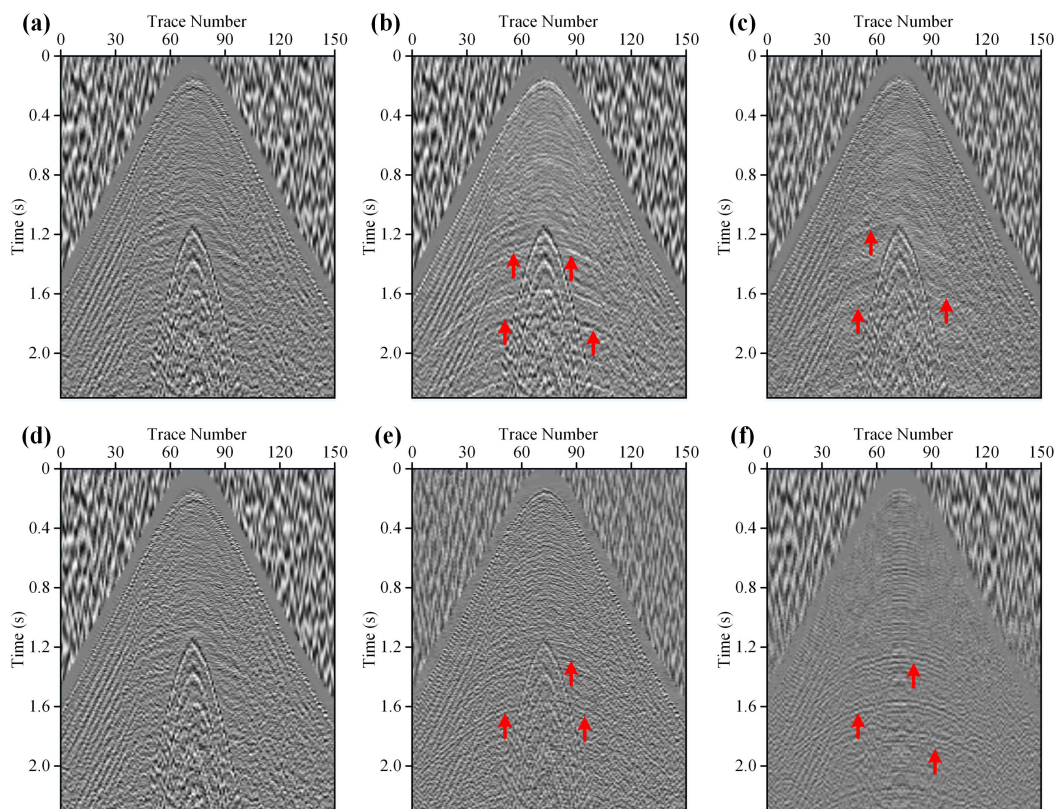


Fig. 11. Comparison of difference records. (a)–(f) Different records of TL-NM, pretrained model, WGN model, target-domain noise model, RPCA, and wavelet transform, respectively.

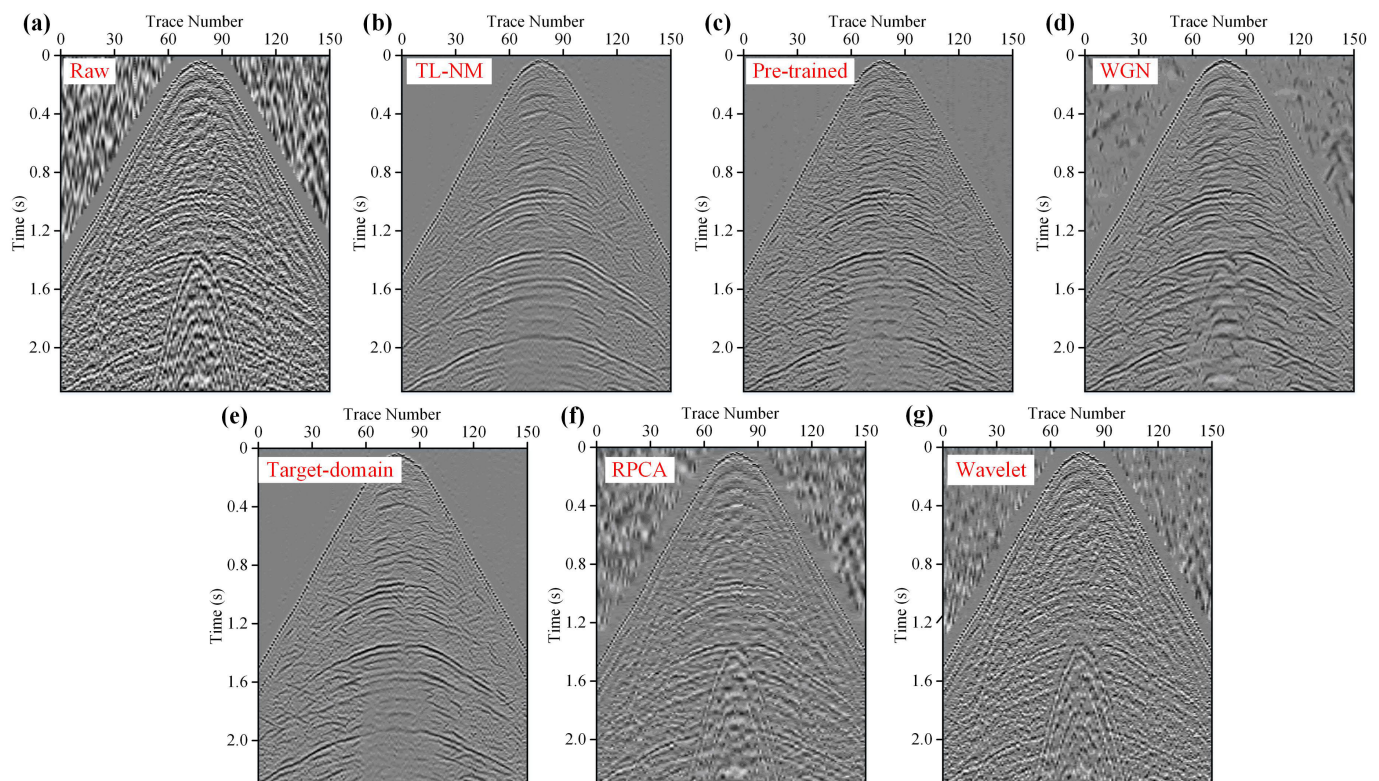


Fig. 12. Another CSP record. (a) Another CSP record extracted from the 3-D shot gather. (b)–(g) Denoised results of Fig. 12(a) by using different methods.

distribution difference between WGN and real seismic random noise. It can be discovered that transfer learning has little effect on the large “distance” between the source domain and

the target domain. Therefore, exploring the high-fidelity noise modeling method is essential for the successful application of transfer learning in seismic data denoising.

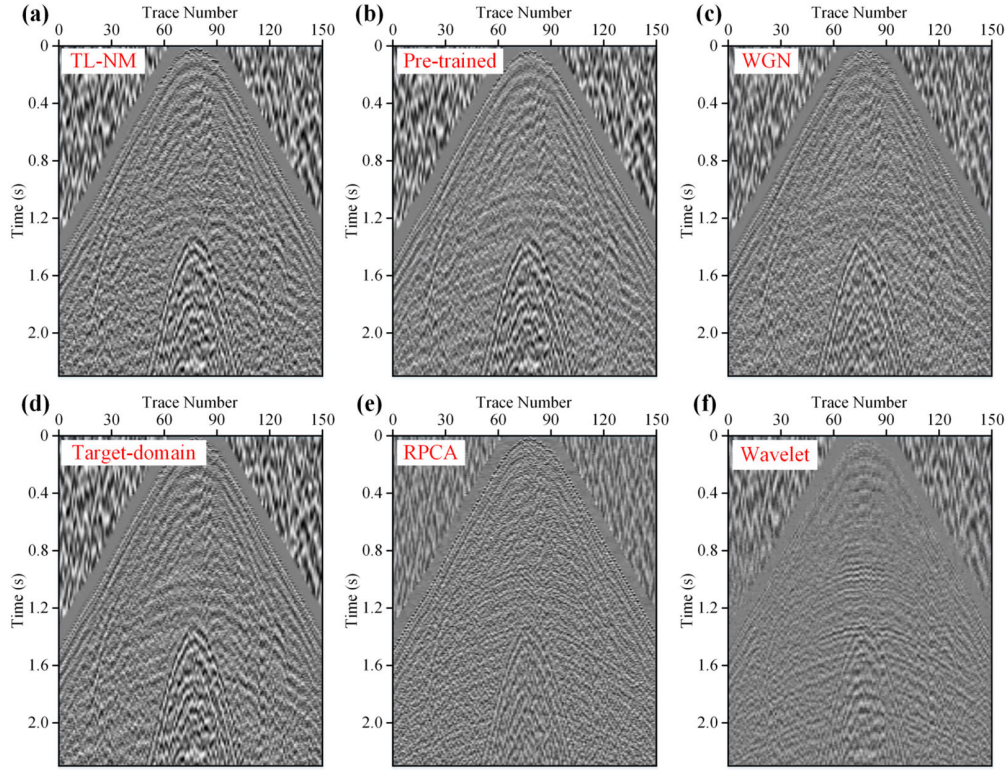


Fig. 13. Different records corresponding to Fig. 12. (a)–(f) Difference records to Fig. 12 (b)–(g), successively.

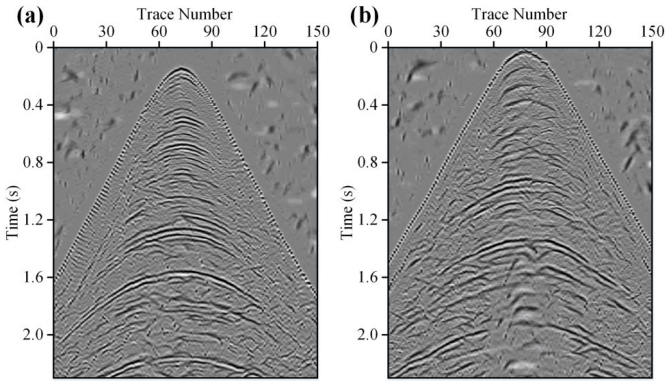


Fig. 14. Effect of synthetic seismic random noise. (a) and (b) Denoised results of Figs. 10(a) and 12(a) by using model 1.

B. Advantage of MCRA-Net

In this article, we compared the proposed network architecture (i.e., MCRA-Net) with two classical architectures of CNN: denoising feedforward CNN (DnCNN) [34] and U-Net [36]. The network depth of DnCNN and U-Net is 22 and 28, respectively. Patch size, batch size, learning rate, epoch number, and convolutional kernel of the two are 64×64 , 128×128 , $[10^{-3}, 10^{-5}]$, 50, and 3×3 , respectively. U-Net performs convolutions on four resolutions: 64×64 , 32×32 , 16×16 , and 8×8 . We leverage the 25 000 signal patches and 25 000 target-domain noise patches to train DnCNN and U-Net. Afterward, the two trained models are adopted

to process the two real CSP records shown in Figs. 10(a) and 12(a). Fig. 15(a)–(d) shows the denoised results and difference records of Fig. 10(a) after applying DnCNN and U-Net; Fig. 15(e)–(h) corresponds to Fig. 12(a). As shown in Fig. 15(a), (c), (e), and (g), some weak events indicated by red arrows show poor continuity, demonstrating that the signal recovery ability of DnCNN and U-Net is worse than that of MCRA-Net; observable signal leakages in Fig. 15(b), (d), (f), and (h) also demonstrate this fact. Furthermore, we can discover some residual random noise in the denoised results by using U-Net [see Fig. 15(c) and (g)]. This experiment suggests that multiresolution features can enhance the performance of CNN-based seismic denoising methods.

C. *t*-SNE

The premise of TL-NM's good denoising performance is that the probability distribution of synthetic noise generated by the Green-function-based noise modeling method is similar to that of target-domain noise. In this part, we utilize *t*-distributed stochastic neighbor embedding (*t*-SNE) [42] to visually compare the probability distribution of synthetic noise data and real noise data. We select 1000 data patches from the 25 000 synthetic noise patches for pretraining and the 1500 real noise patches for fine-tuning. Afterward, the *t*-SNE visualization is performed on the 2000 selected noise patches and the result is shown in Fig. 16. The obvious overlap demonstrates that the probability distributions of synthetic noise and target-domain noise are relatively similar.

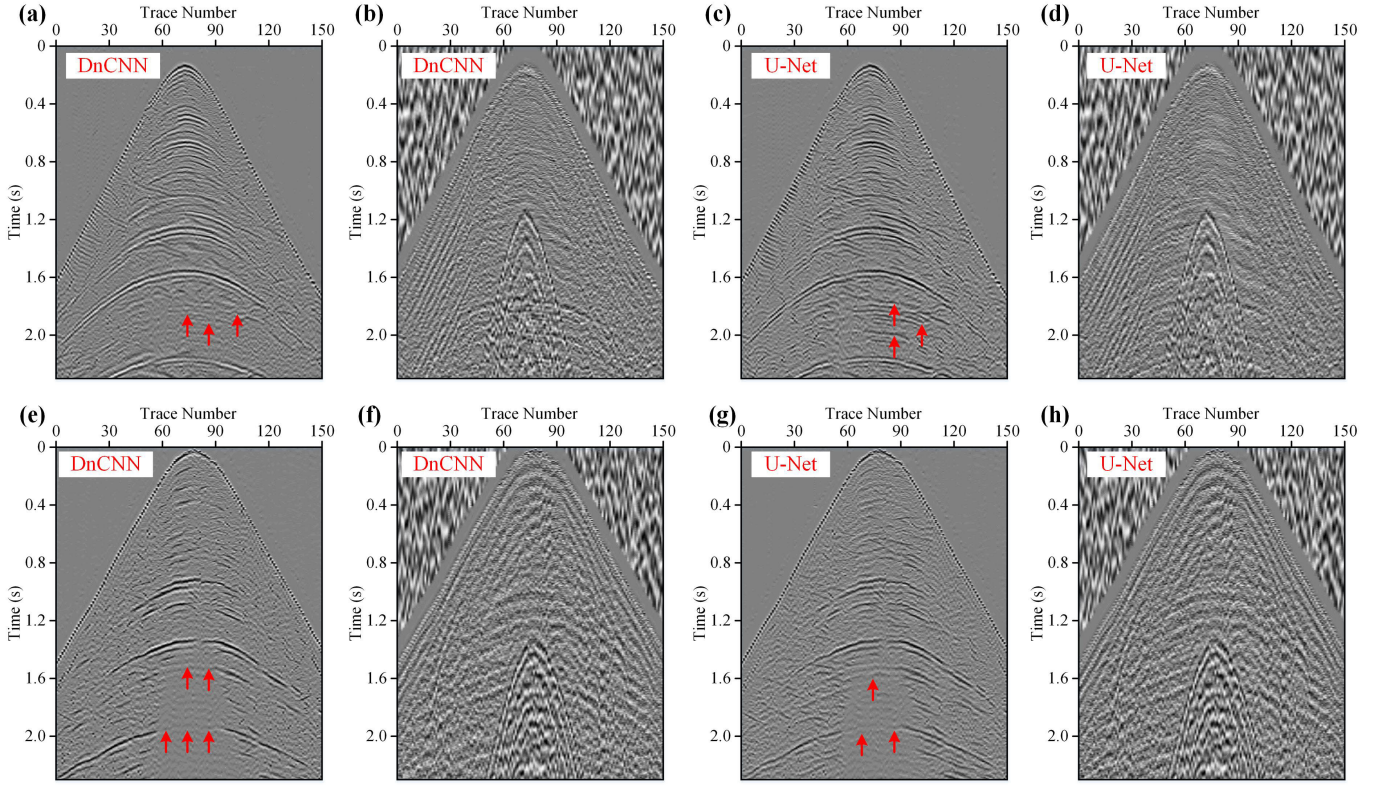


Fig. 15. (a) Denoised result of Fig. 10(a) after applying DnCNN and (b) is its difference record. (c) Denoised result of Fig. 10(a) after applying U-Net and (d) is its difference record. (e) Denoised result of Fig. 12(a) after applying DnCNN and (f) is its difference record. (g) Denoised result of Fig. 12(a) after applying U-Net and (h) is its difference record.

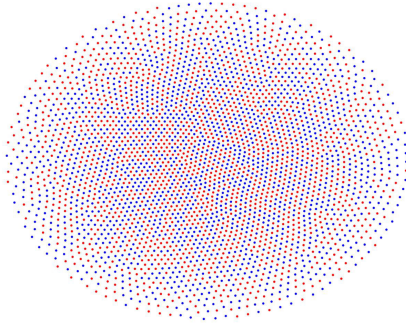


Fig. 16. Visualization of synthetic and real noise patches by using t-SNE. Red and blue dots represent synthetic and real noise patches, respectively.

VII. CONCLUSION AND FUTURE WORK

In this article, we propose a novel DL seismic denoising paradigm, called TL-NM, to mitigate the dependence on target-domain noise data. The nature of TL-NM is transfer learning driven by Green-function-based noise modeling. We utilize a large amount of synthetic noise generated by the Green function to accomplish the pretraining in the source domain, and then, limited target-domain noise data are used to transfer the pretrained model from source domain to target domain. Both in synthetic and real examples, the denoising performance of TL-NM is extremely similar to that of utilizing sufficient target-domain noise data to train the network.

Although TL-NM exhibits considerable potential to solve the problem: lack of target-domain noise data, there are still

three problems to be explored. First, how to generate synthetic noise similar to the real condition is an important basis of TL-NM. Therefore, we need to explore suitable modeling methods for different seismic noise, which is essential for the further application of TL-NM. Second, we utilize synthetic and real random noise to train the MCRA-Net, but the trained model still can suppress the surface waves in real example. This contradictory phenomenon is most likely caused by the obvious velocity difference between the reflected signal and the surface wave. Third, we utilize a relatively simple forward modeling method to generate the signal dataset, which may lead to signal leakage when processing more complex seismic data. How to increase the authenticity of signal dataset is our future work, which is essential for the signal protection ability of TL-NM.

APPENDIX

In this appendix, we review the adopted noise modeling method based on the Green function. This method considers the seismic random noise as the superposition of wind noise, near-field noise, and far-field noise.

Based on the Shinozuka theory [43], the fluctuating wind velocity $v_f(t)$ can be expressed as

$$v_f(t) = \sum_{i=1}^n \sqrt{2S(\omega_i)\Delta\omega} \cos(\omega_i t + \varepsilon_i) \quad (5)$$

where n is the number of components, $S(\omega_i)$ denotes the wind speed spectrum, ω_i represents the frequency of the

i th component, ε_i is the initial phase of the i th component, and $\Delta\omega$ stands for the frequency increment. As shown in the following equation, the wind velocity v is the sum of average wind velocity $\bar{v}$ and v_f :

$$v = \bar{v} + v_f. \quad (6)$$

The force of wind on the ground F can be expressed as

$$F = \frac{1}{2}C\rho v^2 S \quad (7)$$

where C is the coefficient of force, ρ stands for the density of air, and S denotes the area of stressed part. F is the source of wave equation, and then, we can generate the wind noise record by solving the wave equation.

For near-field noise, we utilize the pseudo harmonic-wave function $n(t)$ shown in the following equation to act as the source:

$$n(t) = Ae^{-\delta(t-d)^2} \cos[2\pi f_c(t-d)] \quad (8)$$

where A represents the amplitude, d is the time delay, δ stands for modulation coefficient, and f_c is the central frequency.

The source function of far-field noise is shown in the following equation:

$$X(t) = \sum_{l=1}^m a_l \cos(k_l x_l - \omega_l t + \varepsilon_l) \quad (9)$$

where a_l represents the amplitude of the l th component, k_l is the wavenumber of the l th component, ω_l is the frequency of the l th component, and ε_l is the initial phase of the l th component.

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Hongzhou Wang received the B.S. and M.S. degrees from the Department of Communication Engineering, Jilin University, Changchun, China, in 2018 and 2021, respectively, where he is currently pursuing the Ph.D. degree with the College of Instrument Science and Electrical Engineering.

His main research interests include deep learning, image processing, weak signal detection, seismic record denoising, and seismic inversion.



Guanghui Li received the B.S. and M.S. degrees from the Chengdu University of Technology, Chengdu, China, in 2008 and 2012, respectively, and the Ph.D. degree from the School of Communication and Engineering, Jilin University, Changchun, China, in 2016.

She is currently an Associate Professor with Shanxi University, Taiyuan, China. Her research interests include seismic record denoising and reconstruction, weak reflection events detection, and time–frequency signal processing.



Xintong Dong received the B.S. and Ph.D. degrees from the College of Communication Engineering, Jilin University, Changchun, China, in 2016 and 2021, respectively.

He is currently an Associate Professor with the College of Instrumentation and Electrical Engineering, Jilin University. His main research interests include seismic noise attenuation, seismic inversion, migration imaging, full waveform inversion (FWI), deep learning, sparse representation, and microseismic and seismic data reconstruction.



Jun Lin received the B.S. and M.S. degrees in applied geophysics from the Changchun College of Geology, Changchun, China, in 1982 and 1987, respectively.

He is currently a Professor with the College of Instrumentation and Electrical Engineering, Jilin University, Changchun, where he also serves as the Director of the National Geophysical Exploration Equipment Engineering Research Center. He is an Academician of the Chinese Academy of Engineering, Beijing, China. He actively conducts research

in geophysical exploration theory, technologies, and equipment.



Ming Cheng (Graduate Student Member, IEEE) received the B.S. degree in smart grid information engineering and the M.S. degree in communication engineering from Northeast Electric Power University, Jilin City, China, in 2019 and 2022, respectively. She is currently pursuing the Ph.D. degree with the College of Instrument Science and Electrical Engineering, Jilin University, Changchun, China.

Her main research interests include deep learning, image processing, seismic record denoising, and seismic reconstruction.



Shaoping Lu received the B.S. degree in physics from the University of Science and Technology of China (USTC), Hefei, China, in 2004, and the Ph.D. degree in physics from The University of Texas at Austin (UT-Austin), Austin, TX, USA, in 2010.

He was a Research Geophysicist and a Sr. Research Geophysicist at Petroleum Geo-Services, Inc. (PGS), Houston, TX, USA, from 2010 to 2018. He is currently a Professor with the School of Earth Science and Engineering, Sun Yat-sen University (SYSU), Guangzhou, China. His research interests include seismic data processing and imaging for exploration geophysics.